

Susceptible host dynamics explain pathogen resilience to perturbations

Sang Woo Park^{1,2,*} Bjarke Frost Nielsen³ Emily Howerton³ Bryan T. Grenfell^{3,4,5}
Sarah Cobey¹

1 Department of Ecology and Evolution, University of Chicago, Chicago, IL, USA

2 School of Biological Sciences, Seoul National University, Seoul, Republic of Korea

3 Department of Ecology and Evolutionary Biology, Princeton University,

Princeton, NJ, USA

4 High Meadows Environmental Institute, Princeton University, Princeton, NJ,

USA

5 Princeton School of Public and International Affairs, Princeton, NJ, USA

*Corresponding author: sangwoopark@snu.ac.kr

Abstract

Interventions to slow the spread of SARS-CoV-2 significantly disrupted the transmission of other pathogens. As interventions lifted, whether and when human pathogens would eventually return to their pre-pandemic dynamics remains to be answered. Here, we present a framework for estimating pathogen resilience based on how fast epidemic patterns return to their pre-pandemic dynamics. By analyzing time series data from Hong Kong, Canada, Korea, and the US, we quantify the resilience of common respiratory pathogens and further predict when each pathogen will eventually return to its pre-pandemic dynamics. Our predictions are able to distinguish which pathogens should have returned already, and deviations from these predictions reveal long-term impacts of pandemic perturbations. We find a faster rate of susceptible replenishment underlies pathogen resilience and sensitivity to both large and small perturbations. Overall, our analysis highlights the persistent nature of common respiratory pathogens compared to vaccine-preventable infections, such as measles.

Significance Statement

COVID-19 interventions slowed the transmission of many respiratory pathogens in different ways, raising questions about mechanisms driving the variation in responses to intervention measures. To address this gap, we characterized the sensitivity of pathogen transmission to perturbations by quantifying how fast each pathogen returned to its pre-pandemic circulation patterns. We analyzed data from Hong Kong, Canada, Korea, and the US, and showed that common respiratory pathogens are far

37 less sensitive to perturbations than measles, a vaccine-preventable infection. Finally,
38 we showed that the speed of replenishment of the susceptible population—for exam-
39 ple, through waning immunity—largely determines this sensitivity, suggesting that
40 persistence of common respiratory pathogens is likely driven by rapid susceptible
41 replenishment.

Introduction

Non-pharmaceutical interventions to slow the spread of SARS-CoV-2 disrupted the transmission of other human respiratory pathogens, adding uncertainties to their future epidemic dynamics and their public health burden [1]. As interventions lifted, large differences in outbreak dynamics were observed across different pathogens in different countries, with some pathogens exhibiting earlier and faster resurgences than others [2, 3, 4]. These differences likely reflect variation in intervention patterns, pathogen characteristics, immigration/importation from other countries, and pre-pandemic pathogen dynamics [5]. Therefore, comparing the differential impact of the pandemic perturbations across pathogens can provide unique opportunities to learn about underlying pathogen characteristics across different populations, such as their transmissibility or duration of immunity [6].

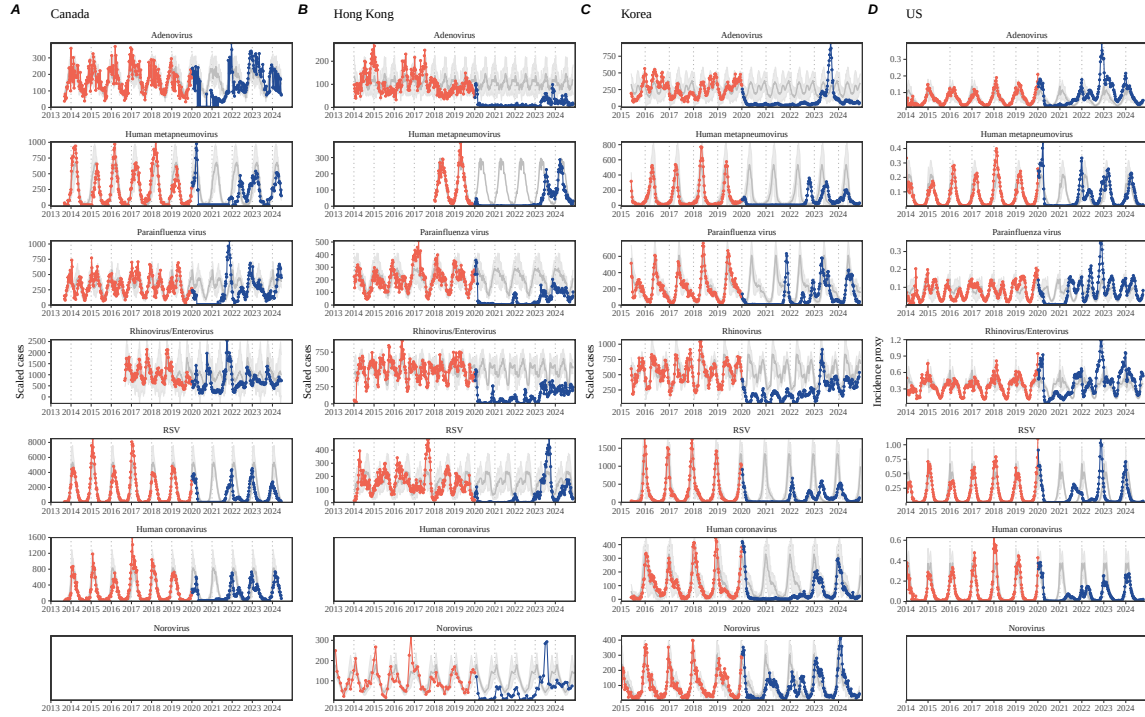


Figure 1: **Observed heterogeneity in responses to pandemic perturbations across respiratory pathogens and norovirus in (A) Canada, (B) Hong Kong, (C) Korea, and (D) US.** Red points and lines represent data before 2020. Blue points and lines represent data since 2020. Gray lines and shaded regions represent the mean seasonal patterns and corresponding 95% confidence intervals around the mean. Mean seasonal patterns were calculated by aggregating cases before 2020 into 52 weekly bins and taking the average in each week. Cases were scaled to account for changes in testing patterns (Materials and Methods).

Even though more than five years have passed since the emergence of SARS-CoV-

2, we still observe persistent changes in pathogen dynamics following the pandemic perturbations. For example, compared to pre-pandemic, seasonal patterns, human metapneumovirus in Korea seems to circulate at lower levels, whereas RSV in Korea seems to exhibit different seasonality (Figure 1). These observations suggest the possibility of a long-term change in pathogen dynamics following the pandemic perturbations, which might be driven by a long-term shift in human behavior or population-level immunity [7, 8]. For example, the emergence of SARS-CoV-2 could have caused a long-term shift in population-level immunity through its interactions with other pathogens [9], especially with seasonal coronaviruses [7, 10, 11]. The possibility of a long-lasting impact of the pandemic perturbations poses an important question for future infectious disease dynamics: can we predict whether and when other pathogens will eventually return to their pre-pandemic dynamics?

So far, most analyses of respiratory pathogens after pandemic perturbations have focused on characterizing the timing of rebound (i.e., when the incidence first returns above historical thresholds) and linking this timing to the relaxation of intervention measures or changes in population susceptibility [1, 12, 5]. While quantifying this timing is particularly useful for public health agencies as it allows them to prepare for future outbreaks, epidemic rebound differs from the recovery of the underlying dynamics—the rebound marks the initial return of pathogen activity without providing any information about the subsequent dynamics. Instead, we seek to characterize how fast (and whether) a pathogen returns to its pre-pandemic dynamics by utilizing all of dynamical information following the perturbation, rather than simply focusing on the initial rebound. These two concepts have a subtle but important difference. For example, it took more than 3 years for human metapneumovirus to rebound in Hong Kong, but the observed epidemic patterns in 2024 appear similar to the pre-pandemic seasonal mean, suggesting a potential, rapid return to pre-pandemic dynamics within 2 years (Figure 1). Measuring this rate of return is useful because it allows us to quantify the ecological resilience of a host-pathogen system, which can inform responses to future interventions [13, 14, 15, 16].

In this study, we lay out theoretical and statistical approaches to characterizing the resilience of a host-pathogen system based on how fast the system recovers from perturbation. We begin by laying out a few representative scenarios that capture the potential impact of pandemic perturbations on endemic pathogen dynamics and illustrate how resilience can be measured by comparing the pre- and post-pandemic dynamics of susceptible and infected hosts. In practice, information on susceptible hosts is often unavailable, making this theoretical approach infeasible. Instead, we utilize a mathematical technique to reconstruct attractors from the data [17], which allows us to measure the rate at which the host-pathogen system approaches this empirical attractor after a perturbation; we define this rate to be the empirical resilience of the host-pathogen system. We use this method to analyze pathogen surveillance data for respiratory and non-respiratory pathogens from Canada, Hong Kong, Korea, and the US. Finally, we show that susceptible host dynamics explain variation in pathogen resilience and demonstrate that more resilient pathogens will be

less sensitive to perturbations caused by demographic stochasticity, thereby providing a direct link between pathogen resilience and persistence.

Conceptual introduction to pathogen resilience

In the classical ecological literature, the resilience of an ecological system is measured by the rate at which the system is expected to return to its reference state following a perturbation [13, 14, 15, 16]. This rate corresponds to the largest real part of the eigenvalues of the linearized system near equilibrium—here, we refer to this value as the *intrinsic* resilience of the system. In practice, we rarely know the true, underlying data generating process that describes the dynamics of common respiratory pathogens, which can depend on many hidden variables, such as changes in population-level susceptibility. Instead, we have to rely on simplifying model assumptions, raising questions about whether we can infer the intrinsic resilience of a system without measurements of all variables that describe the observed dynamics. Nonetheless, we can measure the *empirical* resilience of a host-pathogen system by looking at how fast the system returns to the pre-perturbation endemic dynamics after the perturbation has ended. The COVID-19 pandemic provides a crucial example of a major perturbation, providing unique opportunities to measure the resilience of a host-pathogen system across different countries. As we show later, this measure of resilience depends on the rate of susceptible replenishment, where faster replenishment (e.g., through faster waning immunity) causes a pathogen to be more resilient, and further determines how the system responds to small, stochastic perturbations. Therefore, quantifying pathogen resilience provides key information about their dynamical characteristics.

Resilience of a single-strain system under a short-term perturbation.

As an example, we begin with a simple Susceptible-Infected-Recovered-Susceptible (SIRS) model with seasonally forced transmission and demography (i.e., birth and death) [18]. Here, a pandemic perturbation that reduces transmission by 50% for 6 months starting in 2020 causes the epidemic to deviate from its original stable annual cycle and eventually return (Figure 2A). To quantify resilience, we measure the distance from the pre-pandemic attractor—a set of points in state space or phase plane that the system is pulled towards [19]—by taking Euclidean distance on the SI phase plane relative to the counterfactual unperturbed trajectory (Figure 2B; Materials and Methods). This distance decreases exponentially with time on average, as demonstrated by the locally estimated scatterplot smoothing (LOESS) fit, and the overall rate of return matches the intrinsic resilience of the seasonally unforced system (Figure 2C).

Resilience of a single-strain system under a long-term perturbation.

Alternatively, pandemic perturbations can have a lasting impact on the forces driving pathogen dynamics through a long-term reduction in transmission or permanent change in immunity. For example, if a 10% reduction in transmission persists even

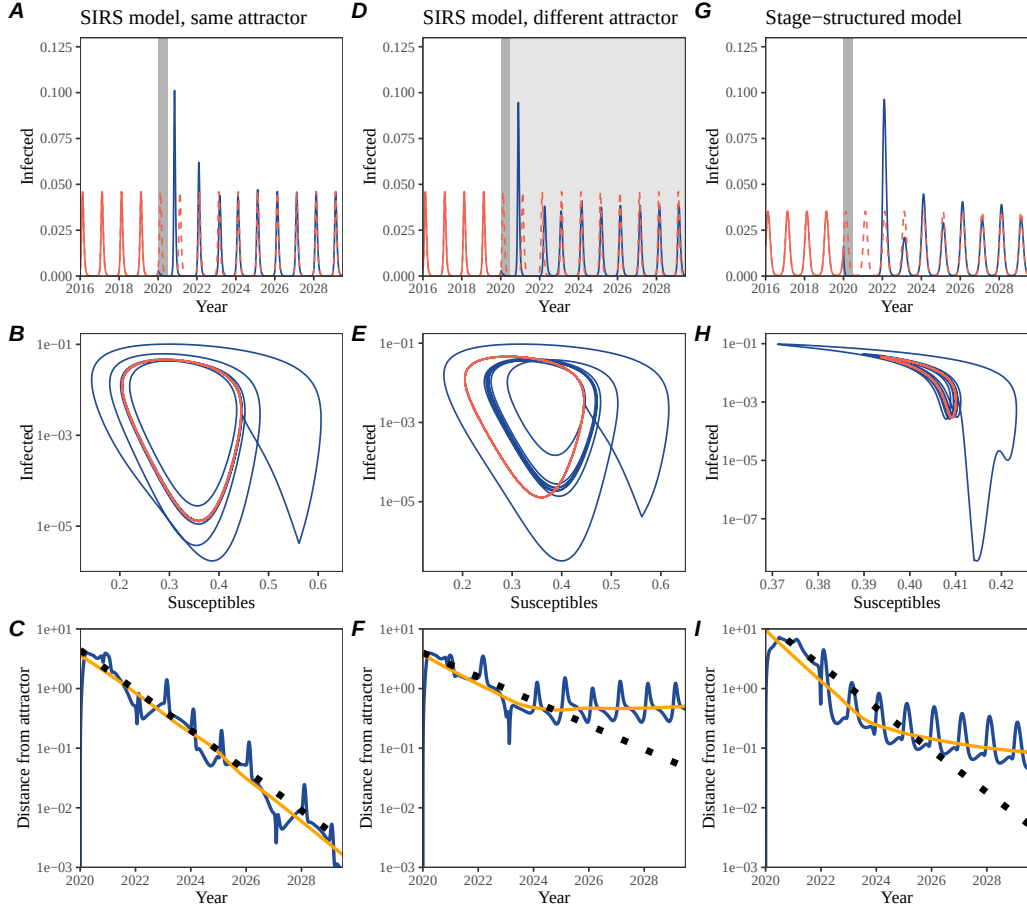


Figure 2: **A simple method to measure pathogen resilience following pandemic perturbations across different scenarios.** (A, D, G) Simulated epidemic trajectories across various models. Red and blue solid lines represent epidemic dynamics before and after pandemic perturbations are introduced, respectively. Red dashed lines represent counterfactual epidemic dynamics in the absence of perturbations. Gray regions indicate the duration of perturbations with the dark gray region representing a 50% transmission reduction and the light gray region representing a 10% transmission reduction. (B, E, H) Phase plane representation of the time series in panels A, D, and G alongside the corresponding susceptible host dynamics. Red and blue solid lines represent epidemic trajectories on an SI phase plane before and after perturbations are introduced, respectively. (C, F, I) Changes in distance from the attractor over time on a log scale. Blue lines represent the distance from the attractor. Orange lines represent the locally estimated scatterplot smoothing (LOESS) fits to the logged distance from the attractor. Dotted lines show the intrinsic resilience of the seasonally unforced system.

138 after the major pandemic perturbations are lifted, the system will converge to a new
 139 attractor (Figure 2D–F), but we may not be able to confirm the convergence to a
 140 new attractor in practice until many years have passed. Nonetheless, we can still
 141 measure the distance from the pre-pandemic attractor (Figure 2E), which initially
 142 decreases exponentially on average (equivalently, linearly on a log scale) and even-
 143 tually plateaus (Figure 2F). Here, a permanent 10% reduction in transmission rate
 144 slows the system, causing a slower decrease in the distance from the attractor (Figure
 145 2F) than it would have otherwise (Figure 2C). This example shows that resilience
 146 is not necessarily an intrinsic property of a specific pathogen. Instead, pathogen re-
 147 siliance is a property of a specific attractor that a host-pathogen system approaches,
 148 which depends on both pathogen and host characteristics.

149 **Resilience of a single-strain system with long-term transients.** Finally,
 150 transient phenomena can further complicate the picture (Figure 2G–I). For exam-
 151 ple, a stage-structured model that accounts for reduction in secondary susceptibility
 152 initially exhibits a stable annual cycle, but perturbations from a 50% reduction in
 153 transmission for 6 months cause the epidemic to shift to a transient biennial cycle
 154 before it eventually approaches the original pre-pandemic attractor (Figure 2G,H).
 155 As indicated by the LOESS fit, the distance from the attractor initially decreases
 156 exponentially at a rate that is consistent with the intrinsic resilience of the seasonally
 157 unforced stage-structured system, but the rate of return slows down with the damped
 158 oscillations (Figure 2I). This behavior is also referred to as a ghost attractor, which
 159 causes long transient dynamics and slow transitions [20]. Strong seasonal forcing in
 160 transmission can also lead to transient phenomena for a simple SIRS model, causing
 161 a slow return to pre-perturbation dynamics (Supplementary Figure S1). However,
 162 we note that a small amount of importation of infections can cause this transient to
 163 disappear and cause the system to become much more resilient, even though it has
 164 negligible effect on overall dynamics or their empirical resilience under non-seasonal
 165 or moderately seasonal transmission (Supplementary Figure S2); under non-seasonal
 166 transmission, importation has limited effect on overall resilience of the system unless
 167 the amount of importation is increased to a unrealistically high value (Supplementary
 168 Figure S3).

169 **Resilience of a two-strain system.** This empirical approach allows us to
 170 measure the resilience of a two-strain host-pathogen system as well even when we
 171 have incomplete observation of the infection dynamics. Simulations from a simple
 172 two-strain competition system illustrate that separate analyses of individual strain
 173 dynamics (e.g., RSV subtype A vs B) and a joint analysis of total infections (e.g.,
 174 total RSV infections) yield identical resilience estimates (Supplementary Figure S4,
 175 5). This is expected because eigenvalues determine the dynamics of the entire system
 176 around the equilibrium, meaning that both strains should exhibit identical rates of
 177 return following a perturbation. Analogous to a single-strain system, strong sea-
 178 sonal forcing in transmission can cause the two-strain system to slow down through
 179 transient phenomena (Supplementary Figure S6).

180 These observations yield three insights. First, we can directly estimate the empir-

ical resilience of a host-pathogen system by measuring the rate at which the system approaches an attractor. The empirical approach to estimating pathogen resilience is particularly convenient because it does not require us to know the true, underlying data generating process; estimating the intrinsic resilience from fitting misspecified models can lead to biased estimates (Supplementary Figure S7). Second, resilience estimates allow us to make phenomenological predictions about the dynamics of a host-pathogen system following a perturbation. Assuming that an attractor has not changed and the distance from the attractor will decrease exponentially over time, we can estimate when the system should reach an attractor. Finally, a change in the (exponential) rate of approach can provide information about whether the system has reached an alternative attractor, or a ghost attractor, that is different from the original, pre-pandemic attractor. These alternative attractors may reflect permanent changes in transmission patterns as well as changes in immune landscapes. Now that several years have passed since major interventions have been lifted, many respiratory pathogens may have had sufficient time to begin returning to their post-intervention attractors—empirical comparisons between current cases and pre-pandemic cases are consistent with this hypothesis (Figure 1). With recent data, we can start to evaluate whether we see early signs of convergence to the former attractor or a new one.

Inferring pathogen resilience from real data

Based on these patterns, we now lay out our approach to estimating pathogen resilience from real data (Figure 3). We first tested this approach against simulations and analyzed case time series of respiratory pathogens from four countries: Canada, Hong Kong, Korea, and the US.

So far, we have focused on simple examples that assume a constant transmission reduction during the pandemic. However, in practice, the impact of pandemic perturbations on pathogen transmission is likely more complex (Figure 3A), reflecting the introduction and relaxation of various intervention strategies. For example, continued transmission reduction from interventions can limit the size of the first outbreak following the rebound, allowing for a larger outbreak in when interventions are further relaxed (Figure 3B).

Previously, we relied on the dynamics of susceptible and infected hosts to compute the distance from the attractor (Figure 2), but information on susceptible hosts is rarely available in practice. In addition, uncertainties in case counts due to observation error, strain evolution, and multiannual cycles in the observed epidemic dynamics (e.g., adenovirus circulation patterns in Hong Kong and Korea) add challenges to defining pre-pandemic attractors. To address these challenges, we can reconstruct an empirical attractor by utilizing Takens’ theorem [17], which states that an attractor of a nonlinear multidimensional system can be mapped onto a delayed embedding (Materials and Methods). Specifically, by taking the logged values of pre-pandemic cases $\log(C(t) + 1)$ and creating delayed copies, Takens’ theorem allows us to define

the pre-pandemic attractor as a points on an M -dimensional space:

$$\langle \log(C(t) + 1), \log(C(t - \tau) + 1), \dots, \log(C(t - (M - 1)\tau) + 1) \rangle, \quad (1)$$

where the delay τ and embedding dimension M are determined based on autocorrelations and false nearest neighbors, respectively [21, 22]. Intuitively, the time series of case counts is generated from a complex system that depends on many hidden variables, such as changes in susceptibility. Takens' theorem tells us that, delayed copies of the observed series preserves the dynamics of the underlying multi-dimensional attractor. We can then apply the same delay and embedding dimensions to the entire time series to determine the position in multi-dimensional state space (Figure 3D) and measure the nearest neighbor distance between the current state of the system and the empirical pre-pandemic attractor (Figure 3E). In theory, the rate at which the distance from this attractor decreases, which can be estimated by fitting a linear regression on a log scale, provides an empirical measure of pathogen resilience (Figure 3E). However, resulting estimates of pathogen resilience can be sensitive to choices about embedding delays and dimensions, where longer delays and higher dimensions tends to smooth out temporal variations in the distance from the attractor (Supplementary Figure S8).

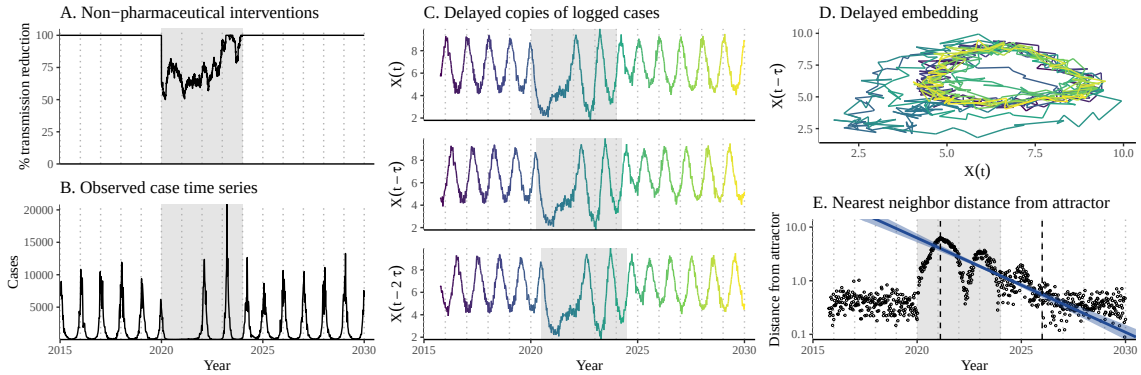


Figure 3: A schematic diagram explaining the inference of pathogen resilience from synthetic data. (A) A realistic example of simulated pandemic perturbations, represented by a relative reduction in transmission. (B) The impact of the synthetic pandemic perturbation on epidemic dynamics simulated using a SIRS model with demographic stochasticity. (C) Generating delayed copies of the logged time series allows us to obtain an embedding. (D) Two dimensional representation of an embedding. (E) Delayed embedding allows us to calculate the nearest neighbor distance from the empirical attractor, which is determined based on the pre-pandemic time series. Blue lines and dashed regions represent the linear regression fit and the associated 95% confidence interval. This distance time series can be used to infer pathogen resilience after choosing an appropriate window for linear regression. For illustration purposes, an arbitrary fitting window was chosen.

We note that estimating pathogen resilience from linear regression is sensitive

to choices about regression fitting windows (Figure 3E). Therefore, we developed an automated window selection criteria for linear regression and tested it against randomized, stochastic simulations across a range of realistic pandemic perturbation shapes. We also explored optimal choices for embedding dimensions and fitting window parameters by quantifying correlation coefficients between the estimated resilience and the intrinsic resilience of a seasonally unforced system (Materials and Methods). Estimation performances varied widely (correlation coefficients ranging from 0.21 to 0.61), but the automated approach consistently outperformed a naive approach, which relies on all data from peak distance to the present (Supplementary Figure S9).

We then applied the best-performing window selection criteria and embedding dimension to pathogen surveillance data presented in Figure 1 (Materials and Methods). For each time series, we applied Takens’ theorem independently to reconstruct the empirical attractor and obtained the corresponding time series of distances from attractors (Supplementary Figure S10). Then, we fitted a linear regression and estimated the empirical resilience for each pathogen in each country (Supplementary Figure S10). The window selection criteria gave poor regression windows in three cases (norovirus in Hong Kong and Korea and rhinovirus/enterovirus in the US), leading to unrealistically low resilience estimates, and so we used ad-hoc regression windows instead (Supplementary Figure S11; Materials and Methods).

For all pathogens we considered, resilience estimates fall between 0.4/year and 1.8/year (Figure 4A). We estimated the mean resilience of common respiratory pathogens to be 0.99/year (95% CI: 0.81/year–1.18/year). For reference, this is ≈ 7.5 times higher than the intrinsic resilience of pre-vaccination measles in England and Wales (≈ 0.13 /year). Finally, resilience estimates for norovirus, a gastrointestinal pathogen, were comparable to those of common respiratory pathogens: 1.44/year (95% CI: 1.01/year–1.87/year) for Hong Kong and 1.07/year (95% CI: 0.86/year–1.29/year) for Korea. Based on a simple ANOVA test, we did not find significant differences in resilience estimates across countries ($p = 0.25$) or pathogens ($p = 0.67$).

Using resilience estimates, we predicted when each pathogen would hypothetically return to their pre-pandemic dynamics, assuming no long-term change in the attractor. Specifically, we extended our linear regression fits to distance-from-attractor time series and ask when the predicted regression line will cross a threshold value; since we relied on nearest neighbor distances, pre-pandemic distances are always greater than zero (Figure 3E), meaning that we can use the mean of pre-pandemic distances as our threshold.

We predicted that a return to pre-pandemic cycles has occurred or would be imminent for most pathogens (Figure 4B). In particular, we predicted that 12 out of 23 pathogen-country pairs should have already returned before the end of 2024. For almost all pathogens that were predicted to have returned already, the observed epidemic dynamics showed clear convergence towards their pre-pandemic seasonal averages, confirming our predictions (Figure 4C). However, there were a few exceptions, including norovirus in Hong Kong and rhinovirus/enterovirus in the US,

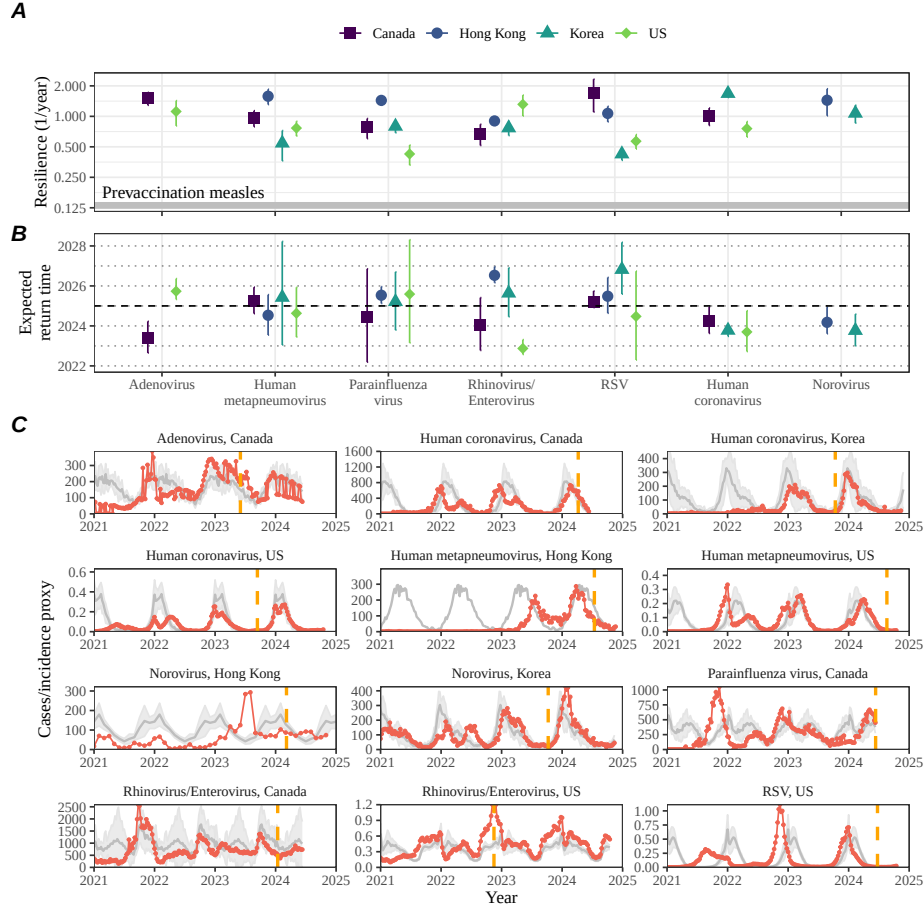


Figure 4: **Summary of resilience estimates and predictions for return time.** (A) Estimated pathogen resilience. The gray horizontal line represents the intrinsic resilience of pre-vaccination measles dynamics. (B) Predicted timing of when each pathogen will return to their pre-pandemic cycles. The dashed line in panel B indicates the end of 2024. Error bars represent 95% confidence intervals. (C) Observed dynamics for pathogen that are predicted to have returned before the end of 2024. Red points and lines represent data before 2020. Gray lines and shaded regions represent the mean seasonal patterns and corresponding 95% confidence intervals around the mean, previously shown in Figure 1. Orange vertical lines represent the predicted timing of return.

281 where the observed epidemic dynamics in 2024 exhibit clear deviation from their pre-
 282 pandemic seasonal averages (Figure 4C; Figure S11). These observations suggest a
 283 possibility that some common respiratory pathogens may have converged to different
 284 attractors or are still exhibiting non-equilibrium dynamics. In contrast, pathogens
 285 that were predicted to have not returned yet also showed clear differences from their
 286 pre-pandemic seasonal averages; as many of these pathogens are predicted to return

in 2025–2026, we may be able to test these predictions in near future (Supplementary Figure S12). Our reconstructions of distance time series and estimates of pathogen resilience and expected return time were generally robust to choices of embedding dimensions (Supplementary Figure S13–14).

We also evaluated the ability of our approach to make out-of-sample predictions by using observations up to the 26th week of 2023 (Supplementary Figure S15–16). This led to greater uncertainty in resilience estimates and corresponding predictions of the expected return time (Supplementary Figure S16A,B). The expected and observed return times exhibit weak positive correlation ($\rho = 0.36$; 95% CI: -0.34–0.81), and 7 out of 10 confidence intervals contain the true value (Supplementary Figure S16C). We found moderate correlations ($\rho = 0.51$; 95% CI: 0.01–0.80) in predictions for the expected return time between partial regression fits, which rely on observations up to the 26th week of 2023, and full regression fits, which rely on the entire data set (Supplementary Figure S16D); this implies consistency in our predictions. However, resilience estimates from partial regression fits and full regression fits were weakly correlated ($\rho = 0.30$; 95% CI: -0.23–0.70), demonstrating challenges in estimating pathogen resilience from limited data (Supplementary Figure S16E).

Susceptible host dynamics explain variation in pathogen resilience

So far, we have focused on quantifying pathogen resilience from the observed patterns of pathogen re-emergence following pandemic perturbations. But what factors determine the resilience of a host-pathogen system? Here, we illustrate how susceptible host dynamics affect pathogen resilience by varying the basic reproduction number $\mathcal{R}_0$ and the duration of immunity of the SIRS model.

An increase in $\mathcal{R}_0$ and a decrease in the duration of immunity leads to a higher resilience value (Figure 5A) and a faster per-capita rate of susceptible replenishment, the ratio between total rate at which new susceptibles enter the population and the equilibrium number of susceptibles, S^* (Figure 5B). Specifically, a higher $\mathcal{R}_0$ drives a faster per-capita susceptible replenishment rate by decreasing S^* . For a simple SIR model that assumes a life-long immunity, pathogen resilience is proportional to the per-capita rate of susceptible replenishment (Materials and Methods). Overall, these observations suggest that a faster per-capita susceptible replenishment rate causes the system to be more resilient.

By taking the average values of empirical resilience for each pathogen, we were able to map each pathogen onto a set of parameters of the SIRS model that are consistent with corresponding resilience estimates (Figure 5A). Across all pathogens we considered, we estimated that the average duration of immunity is likely to be short (< 4 years) across a plausible range of $\mathcal{R}_0$ (< 6). In contrast, there was a large uncertainty in the estimates for susceptible replenishment rates due to a lack of one-to-one correspondence between susceptible replenishment rates and pathogen

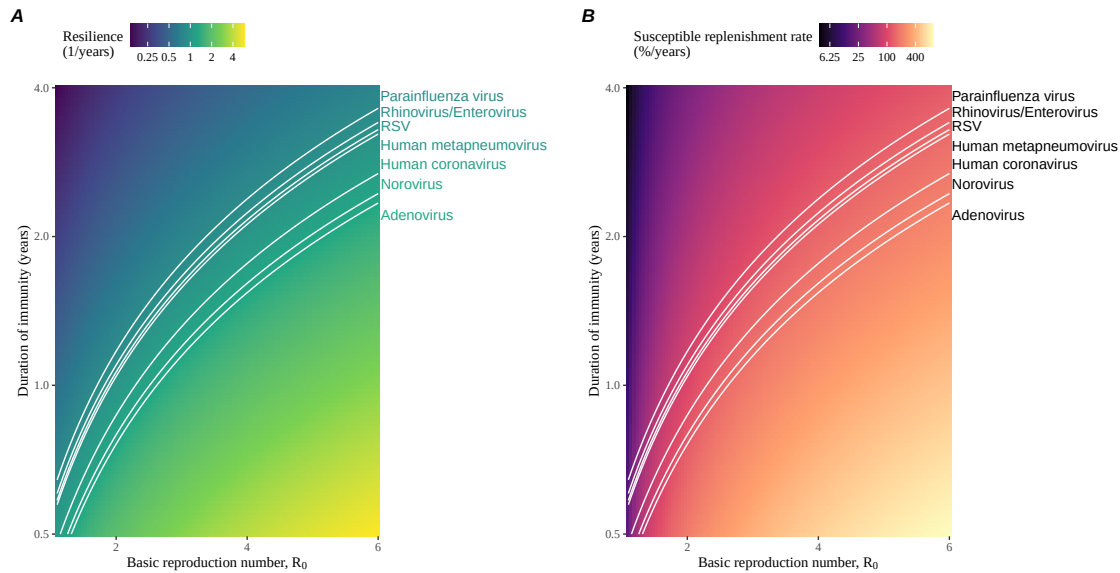


Figure 5: **Linking pathogen resilience to epidemiological parameters and susceptible host dynamics.** (A) Intrinsic resilience as a function of the basic reproduction number $\mathcal{R}_0$ and the duration of immunity. (B) Per-capita susceptible replenishment rate as a function of the basic reproduction number $\mathcal{R}_0$ and the duration of immunity. The standard SIRS model without seasonal forcing is used to compute intrinsic resilience and the per-capita susceptible replenishment rate. Lines correspond to a set of parameters that are consistent with mean resilience estimates for each pathogen, averaged across different countries.

resilience (Figure 5B).

Pathogen resilience determines sensitivity to stochastic perturbations

Even in the absence of major pandemic perturbations, host-pathogen systems are expected to experience continued perturbations of varying degrees from changes in epidemiological conditions, such as human behavior, climate, and viral evolution. These perturbations can also arise from demographic stochasticity, which is inherent to any ecological system. Here, a seasonally unforced SIRS model with constant birth and death rates illustrates that resilience of a host-pathogen system determines the sensitivity to perturbations caused by demographic stochasticity, where a less resilient system experiences a greater deviation from the deterministic trajectory (Figure 6, Materials and Methods). Less resilient systems also exhibited slower epidemic cycles (Figure 6A–C). The periodicity of this epidemic cycle matched those predicted by the intrinsic periodicity of the system (Supplementary Figure S17),

341 which is inversely proportional to the intrinsic resilience (Supplementary Figure S18).
 342 We note that the interplay between seasonal transmission and intrinsic periodicity
 343 can also lead to complex cycles, as illustrated by a recent analysis of *Mycoplasma*
 344 *pneumoniae* dynamics [23].

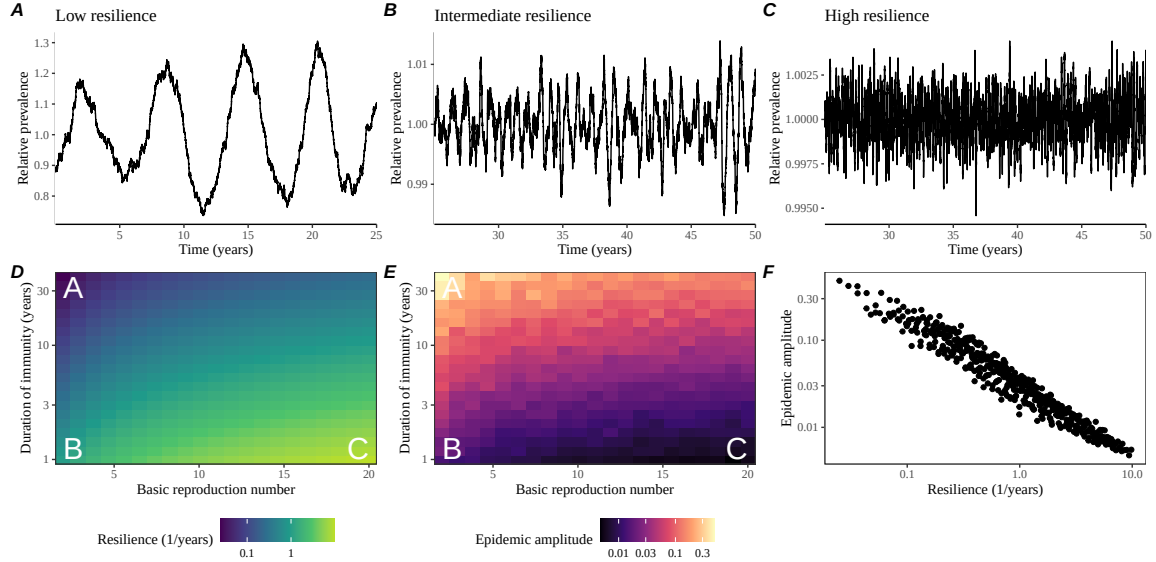


Figure 6: **Linking resilience of a host-pathogen system to its sensitivity to stochastic perturbations.** (A–C) Epidemic trajectories of a stochastic SIRS model across three different resilience values: low, intermediate, and high. The relative prevalence was calculated by dividing infection prevalence by its mean value. (D) Intrinsic resilience of a system as a function of the basic reproduction number $\mathcal{R}_0$ and the duration of immunity. (E) Epidemic amplitude as a function of the basic reproduction number $\mathcal{R}_0$ and the duration of immunity. The epidemic amplitude corresponds to $(\max I - \min I)/(2\bar{I})$, where $\bar{I}$ represents the mean prevalence. Labels A–C in panels D and E correspond to scenarios shown in panels A–C. (F) The relationship between pathogen resilience and epidemic amplitude.

345 We also note that the intrinsic resilience is not the sole determinant for how sen-
 346 sitive the system is to stochastic perturbations. For example, the population size
 347 and average duration of infection also affect the amount of deviation from the deter-
 348 ministic trajectory caused by demographic stochasticity, even though these quantities
 349 have little to no impact on the intrinsic resilience (Supplementary Figure S17). These
 350 conclusions were robust for the seasonally forced SIRS model, where the amount of
 351 deviation from the deterministic trajectory depends on the corresponding resilience
 352 of the seasonally unforced system (Supplementary Figure S19).

Discussion

COVID-19 pandemic interventions caused major disruptions to circulation patterns of both respiratory and non-respiratory pathogens, adding challenges to predicting their future dynamics [1, 2, 3, 4]. These perturbations offer large-scale natural experiments for understanding how different pathogens respond to perturbations. In this study, we showed that pathogen re-emergence patterns following pandemic perturbations can be characterized through the lens of ecological resilience and presented a new method for estimating pathogen resilience from time series data. We showed that variation in pathogen resilience can be explained by the differences in susceptible host dynamics, where faster replenishment of the susceptible pool corresponds to a more resilient host-pathogen system. Finally, we showed that pathogen resilience also determines the sensitivity to stochastic perturbations.

We analyzed case time series of common respiratory infections and norovirus infections from Canada, Hong Kong, Korea, and the US to estimate their resilience. Overall, we estimated the resilience of these pathogens to range from 0.4/year to 1.8/year, which is 3–14 times more resilient than prevaccination measles. Consistent with other epidemiological evidence [24, 25, 26, 27], these resilience estimates indicate that common respiratory pathogens and norovirus likely exhibit faster susceptible replenishment and are therefore more persistent, indicating potential challenges in controlling these pathogens.

Based on our resilience estimates, we made phenomenological predictions about when each pathogen will return to their endemic cycles. Our main regression analysis could accurately distinguish which pathogens should have already returned before the end of 2024. However, there were two main exceptions (i.e., norovirus in Hong Kong and rhinovirus/enterovirus in the US), hinting at long-lasting impact of pandemic perturbations. These changes may reflect changes in surveillance or actual shift in the dynamics. While it may seem unlikely that behavioral changes would only affect a few pathogens and not others, we cannot rule out this possibility given differences in the observed mean age of infections and therefore the differences in age groups that primarily drive transmission [28, 29]. Differences in the mode of transmission between respiratory vs gastrointestinal pathogens may also contribute to the differences in responses to pandemic perturbations.

For almost half of the pathogens we considered, we predicted that their return to original epidemic patterns is imminent. However, our evaluation of out-of-sample predictions suggested major challenges in predicting the exact timing of pathogen return and estimating pathogen resilience from limited data. Therefore, our estimates of prediction timing should be taken as a ballpark estimate. We will need a few more years of data to test whether these pathogens will eventually return to their original dynamics or eventually converge to a different attractor. We also cannot rule out the possibility that some pathogens may exhibit long-term transient behaviors following pandemic perturbations. Overall, these observations echo earlier studies that highlighted the long-lasting impact of pandemic perturbations [8, 30, 31, 4, 23].

395 We showed that susceptible host dynamics shape pathogen resilience, where faster
 396 replenishment of the susceptible population causes the pathogen to be more resilient.
 397 For simplicity, we focused on waning immunity and birth as the main drivers of the
 398 susceptible host dynamics but other mechanisms can also contribute to the replen-
 399 ishment of the susceptible population. In particular, pathogen evolution, especially
 400 the emergence of antigenically novel strains, can cause effective waning of immunity
 401 in the population; therefore, we hypothesize that the rate of antigenic evolution is
 402 likely a key feature of pathogen resilience. Future studies should explore the relation-
 403 ship between the rate of evolution and resilience for antigenically evolving pathogens.
 404 This result also highlights the importance of detailed measurements of changes in the
 405 susceptible population through immune assays for understanding pathogen dynamics
 406 [32].

407 There are many other factors that may affect the resilience of a host-pathogen
 408 system beyond susceptible replenishment rate. For example, we showed that im-
 409 portation can also increase the resilience of the system (Supplementary Figures S2,
 410 3). Interestingly, this effect is minimal without seasonal forcing but substantially
 411 amplified under strong seasonal forcing, suggesting that analyzing the eigenvalues of
 412 seasonally unforced system may be insufficient to capture key aspects of the system’s
 413 resilience. Future studies should consider both analytical and numerical analyses of
 414 more detailed, mechanistic models to understand how other factors—such as the
 415 generation-interval distribution [33], immune boosting [34], strain competition [35],
 416 and post-infection mortality [36]—may alter the resilience of a host-pathogen sys-
 417 tem. Quantifying relative changes in pathogen resilience with respect to underlying
 418 parameters of a model—analogue to sensitivity and elasticity analyses in ecology
 419 [37, 38]—may allow one to infer relative importance of different factors that affect
 420 pathogen resilience.

421 Quantifying pathogen resilience also offers novel approaches to validating population-
 422 level epidemiological models. So far, most model validation in infectious disease ecol-
 423 ogy is based on the ability of a model to reproduce the observed epidemic dynamics
 424 and to predict future dynamics [39, 40, 26, 41, 42]. However, many models can
 425 perform similarly under these criteria. For example, two major RSV models have
 426 been proposed to explain biennial epidemic patterns: (1) a stage- and age-structured
 427 model that allows disease severity to vary with number of past infections and age of
 428 infection [26] and (2) a pathogen-interaction model that accounts for cross immunity
 429 between RSV and human metapneumovirus [40]. Since both models can accurately
 430 reproduce the observed epidemic patterns, standard criteria for model validation
 431 do not allow us to distinguish between these two models from population-level data
 432 alone. Instead, it would be possible to measure the empirical resilience of each model
 433 by simulating various perturbations and comparing the simulations to estimates of
 434 empirical resilience from data, using pandemic perturbations as a reference.

435 There are several limitations to our work. First, we did not extensively explore
 436 other approaches to reconstructing the attractor. Recent studies showed that more
 437 sophisticated approaches, such as using non-uniform embedding, can provide more

robust reconstruction for noisy data [22]. In the context of causal inference, choices about embedding can have major impact on the resulting inference [43]. Our resilience estimates are likely overly confident given a lack of uncertainties in attractor reconstruction as well as the simplicity of our statistical framework. Nonetheless, as illustrated in our sensitivity analyses, inferences about pathogen resilience in our SIRS model appear to be robust to decisions about embedding lags and dimensions—this is because tracking the rate at which the system approaches the attractor is likely a much simpler problem than making inferences about causal directionality. Short pre-pandemic time series also limit our ability to accurately reconstruct the attractor and contribute to the crudeness of our resilience estimates; although this is less likely a problem for respiratory pathogens that are strongly annual, our attractor reconstruction may be inaccurate for those exhibiting multi-annual cycles, such as adenovirus in Hong Kong and Korea. Our framework also does not allow us to distinguish whether a system has settled to a new attractor or is experiencing long-term transient behavior.

Uncertainties in pathogen dynamics due to changes in testing patterns further contribute to the crudeness of our resilience estimates. We tried to account for changes in testing patterns by either scaling the case counts (for Canada, Hong Kong, and Korea) or deriving incidence proxy (for the US). We were unable to derive incidence proxy for Canada and Hong Kong due to instability of testing positivity, including long-term reduction in test positivity associated with an increase in testing volume. Consistency in resilience estimates across different countries provide some robustness to our conclusions, our exact estimates are likely sensitive to changes in testing. We also cannot rule out the possibility that potential long-term impacts of pandemic perturbations may simply reflect changes in testing, rather than actual dynamical changes. Future studies should investigate improved methods for accounting for changes in testing patterns to obtain a more accurate perspective on the differences between current and past outbreak patterns of endemic pathogens.

While attractor reconstruction allows us to make model-free inferences of pathogen resilience, it does not allow us to tease apart how different mechanisms contribute to the resilience a host-pathogen system. Using the simple SIRS model, we illustrated that susceptible host dynamics are key determinants of pathogen resilience, but we also found that there isn’t a one-to-one correspondence between per capita replenishment rate of the susceptible population and pathogen resilience estimates. Future studies should explore using mechanistic models to explain heterogeneity in resilience estimates across different pathogens.

Finally, our simulation-based analyses primarily focused on single-strain systems, but real-world pathogens can interact with other pathogens, which can result in complex dynamics [44, 45]. To address this limitation, we considered a simple model of two competing strains (via cross immunity) and showed that the resilience of a coupled system can be measured by studying the dynamics of either strain. However, this conclusion likely depends on the strength and mechanism of strain interactions. For example, ecological interference between two unrelated pathogens [44] will likely gen-

erate weaker coupling than cross-immunity between related pathogens; in the former case, we do not necessarily expect two unrelated pathogens to have same resilience despite their ecological interference. Some pathogen strains can also exhibit positive interactions where infection by one strain can lead to an increased transmission of another competing strain. For example, previous studies showed that an increased dengue transmission through antibody-dependent enhancement can permit coexistence and persistence of competing strains [46]; since pathogen transmissibility is a major determinant of pathogen resilience, we tentatively hypothesize that positive interactions such as antibody-dependent enhancement may increase the resilience of a system. Future studies should explore how different mechanisms of pathogen interactions contribute to the resilience of each competing pathogen as well as the entire system. Despite these limitations, our study illustrates that quantifying pathogen resilience can provide novel insights into pathogen dynamics. Furthermore, our qualitative prediction that common respiratory pathogens are more resilient than pre-vaccination measles is also likely to be robust, given how rapidly many respiratory pathogens returned to their original cycles following pandemic perturbations.

Predicting the impact of anthropogenic changes on infectious disease dynamics is a fundamental aim of infectious disease research in a rapidly changing world. Our study illustrates that how a host-pathogen system responds to both small and large perturbations is largely predictable through the lens of ecological resilience. In particular, quantifying the resilience of a host-pathogen system offers a unique insight into questions about endemic pathogens' responses to pandemic perturbations, including whether some pathogens will exhibit long-lasting impact from the pandemic perturbation or not. More broadly, a detailed understanding of the determinants of pathogen resilience can provide deeper understanding of pathogen persistence.

Materials and Methods

Data

We gathered time series on respiratory infections from Canada, Hong Kong, Korea, and United States (US). As a reference, we also included norovirus infections when available. In contrast to respiratory pathogens, we hypothesized gastrointestinal viruses, such as norovirus, to be differently affected by pandemic perturbations.

Weekly time series of respiratory infection cases in Canada came from a publicly available website by the Respiratory Virus Detection Surveillance System, which collects data from select laboratories across Canada [47]. Weekly time series of respiratory infection cases in Hong Kong came from a publicly available website by the Centre for Health Protection, Department of Health [48, 49]. Weekly time series of acute respiratory infection cases in Korea came from a publicly available website by the Korea Disease Control and Prevention Agency [50]. Finally, weekly time series of respiratory infection cases in the US were obtained from the National Respiratory and Enteric Virus Surveillance System (NREVSS). Readers can request

the data from NREVSS at nrevss@cdc.gov. Time series on number of tests were also available in Canada, Hong Kong, and the US, but not in Korea.

Data processing

For all time series, we rounded every year to 52 weeks by taking the average number of cases and tests between the 52nd and 53rd week. We also rescaled all time series to account for changes in testing patterns, which were then used for the actual analysis.

For Canada, an increase in testing was observed from 2013 to 2024 (Supplementary Figure S21); this increase is associated with a decrease in test positivity during the same period (Supplementary Figure S22). To account for this increase, we calculated a 2 year moving average for the number of tests for each pathogen, which we used as a proxy for testing effort. Then, we divided the smoothed testing patterns by the smoothed value at the final week such that the testing effort has a maximum of 1. We then divided weekly cases by the testing effort to obtain a scaled case time series. A similar approach was used earlier for an analysis of RSV time series in the US to account for changes in testing patterns [26].

For Hong Kong, we applied the same scaling procedure to the time series as we did for Canada. In this case, we only adjusted for testing efforts up to the end of 2019 because there was a major reduction in testing for common respiratory pathogens between 2020 and 2023 (Supplementary Figure S23). Analogous to patterns in Canada, an increase in testing volume is associated with a decrease in test positivity (Supplementary Figure S24).

For Korea, while we did not have information on testing, the reported number of respiratory infections consistently increased from 2013 to the end of 2019, which we interpreted as changes in testing patterns (Supplementary Figure S25). Since we did not have testing numbers, we used the weekly sum of all acute respiratory viral infection cases as a proxy for testing, which were further smoothed with moving average and scaled to have a maximum of 1. For Korea, we also only adjusted for testing efforts up to the end of 2019.

In the US, there has been a large increase in testing for some respiratory pathogens, especially RSV, which could not be corrected by simple scaling (Supplementary Figure S26). Instead, we derived an incidence proxy by multiplying the test positivity with influenza-like illness positivity, which was taken from <https://gis.cdc.gov/grasp/fluview/fluportaldashboard.html>. This method of estimating an incidence proxy has been recently applied in the analysis of seasonal coronaviruses [7] and *Mycoplasma pneumoniae* infections [4]. Detailed assumptions and justifications are provided in [51].

Data summary

To make qualitative comparisons between pre- and post-perturbation dynamics of respiratory pathogen circulation patterns, we calculated the mean seasonal patterns

560 using time series of either rescaled cases or incidence proxy estimates before 2020. We
 561 did so by taking the mean value in each week across all years before 2020. Confidence
 562 intervals around the means were calculated using a simple t test.

563 Estimating pathogen resilience

564 In order to measure pathogen resilience from surveillance data, we first reconstructed
 565 the empirical pre-pandemic attractor of the system using Takens' embedding theorem
 566 [17]. Specifically, for a given pathogen, we took the pre-pandemic (before 2020) case
 567 time series $C(t)$ and reconstructed the attractor using delayed embedding with a
 568 uniform delay of τ and dimension M :

$$X_{\tau,m}(t) = \langle \log(C(t) + 1), \log(C(t - \tau) + 1), \dots, \log(C(t - (M - 1)\tau) + 1) \rangle. \quad (2)$$

569 Here, the delay τ was determined by calculating the autocorrelation of the logged
 570 pre-pandemic time series and asking when the autocorrelation crosses 0 for the first
 571 time [22]; a typical delay for an annual outbreak is around 13 weeks.

572 Then, for a given delay τ , we determined the embedding dimension M using the
 573 false nearest neighbors approach [21, 22]. To do so, we started with an embedding
 574 dimension e and constructed a set of points $A_{\tau,e} = \{X_{\tau,e}(t) | t < 2020\}$. Then, for
 575 each point $X_{\tau,e}(t)$, we determined the nearest neighbor from the set $A_{\tau,e}$, which we
 576 denote $X_{\tau,e}(t_{nn})$ for $t \neq t_{nn}$. Then, if the distance between these two points in the
 577 $e+1$ dimension, $D_{\tau,e+1}(t) = \|X_{\tau,e+1}(t_{nn}) - X_{\tau,e+1}(t)\|_2$, is larger than their distance in
 578 the e dimension, $D_{\tau,e}(t) = \|X_{\tau,e}(t_{nn}) - X_{\tau,e}(t)\|_2$, these two points are deemed to be
 579 false nearest neighbors; specifically, we used a threshold R for the ratio between two
 580 distances $D_{\tau,e+1}(t)/D_{\tau,e}(t)$ to determine false nearest neighbors. The first embed-
 581 ding dimension e that does not have any false nearest neighbors corresponds to the
 582 embedding dimension M for a given pathogen-country pair. For the main analysis,
 583 we used $R = 10$, which was chosen from a sensitivity analysis against simulated data
 584 (Supplementary Text). Once we determined the embedding lag τ and dimension M ,
 585 we apply the embedding to the entire time series and calculate the nearest neigh-
 586 bor distance against the attractor $A_{\tau,M}$ to obtain a time series of distance from the
 587 attractor $D_{\tau,M}(t)$.

588 From a time series of distances from the attractor, we estimated pathogen re-
 589 siliance by fitting a linear regression to an appropriate window. To automatically
 590 select fitting windows, we began by smoothing the distance time series using locally
 591 estimated scatterplot smoothing (LOESS) to obtain $\hat{D}_{\tau,M}(t)$, where the smoothing
 592 is performed on a log scale and exponentiated afterwards. This smoothing allowed
 593 us to find appropriate threshold values for selecting fitting windows that are insensi-
 594 tive to errors in our estimates of distance from the attractor. Then, we determined
 595 threshold values (T_{start} and T_{end}) for the smoothed distances and choose the fitting
 596 window based on when $\hat{D}_{\tau,M}(t)$ crosses these threshold values for the first time.
 597 These thresholds were determined by first calculating the maximum distance,

$$\max \hat{D} = \max \hat{D}_{\tau,M}(t), \quad (3)$$

598 and the mean pre-pandemic distance,

$$\hat{D}_{\text{mean}} = \frac{1}{N_{t < 2020}} \sum_{t < 2020} \hat{D}_{\tau, M}(t), \quad (4)$$

599 as a reference, and then dividing their ratios into K equal bins,

$$T_{\text{start}} = \hat{D}_{\text{mean}} \times \left(\frac{\max \hat{D}}{\hat{D}_{\text{mean}}} \right)^{(K-a)/K}, \quad (5)$$

$$T_{\text{end}} = \hat{D}_{\text{mean}} \times \left(\frac{\max \hat{D}}{\hat{D}_{\text{mean}}} \right)^{a/K}, \quad (6)$$

600 where a represents the truncation threshold. This allows us to discard the initial
 601 period during which the distance increases (from the introduction of intervention
 602 measures) and the final period during which the distance plateaus (as the system
 603 reaches an attractor). The fitting window is determined based on when the smoothed
 604 distance $\hat{D}_{\tau, M}(t)$ crosses these threshold values for the first time; then, we fit a
 605 linear regression to logged (unsmoothed) distances $\log D_{\tau, M}(t)$ using that window.
 606 Alongside the threshold R for the false nearest neighbors approach, we tested optimal
 607 choices for K and a values using simulations (Supplementary Text). We used $K = 19$
 608 and $a = 2$ throughout the paper based on the simulation results. We excluded data
 609 sets from the regression analysis if the logged average of the estimated distance from
 610 the attractor during the last year is at least 2-fold greater than the logged average
 611 of the pre-pandemic distance from the attractor.

612 To evaluate the ability to predict the precise return time for out-of-sample ob-
 613 servations, we applied the same framework using data up to the 26th week of 2023.
 614 In doing so, we compared the predicted return time with the observed return time,
 615 which was determined based on when the smoothed empirical distance from the
 616 attractor became close enough to pre-pandemic levels. As a pathogen returns to
 617 its pre-pandemic cycle, the the smoothed empirical distance from the attractor is
 618 expected to converge to the pre-pandemic levels $\hat{D}_{\text{mean}}$, but the empirical distance
 619 may not become smaller than $\hat{D}_{\text{mean}}$. Therefore, we used $1.2\hat{D}_{\text{mean}}$ as our thresh-
 620 old to obtain an approximate return time. We note that this approach necessarily
 621 underestimates the observed return time and therefore should be interpreted as a
 622 conservative estimate. For additional validation, we compared these out-of-sample
 623 predictions with return time predictions based on full time series.

624 Mathematical modeling

625 Throughout the paper, we use a series of mathematical models to illustrate the
 626 concept of pathogen resilience and to understand the determinants of pathogen re-
 627 siliance. In general, the intrinsic resilience of a given system is given by the largest
 628 real part of the eigenvalues of the linearized system at endemic equilibrium. Here, we

focus on the SIRS model with demography (birth and death) and present the details of other models in Supplementary Text. The SIRS (Susceptible-Infected-Recovered-Susceptible) model is the simplest model that allows for waning of immunity, where recovered (immune) individuals are assumed to become fully susceptible after an average of $1/\delta$ time period. The dynamics of the SIRS model is described by the following set of differential equations:

$$\frac{dS}{dt} = \mu - \beta(t)SI - \mu S + \delta R \quad (7)$$

$$\frac{dI}{dt} = \beta(t)SI - (\gamma + \mu)I \quad (8)$$

$$\frac{dR}{dt} = \gamma I - (\delta + \mu)R \quad (9)$$

where μ represents the birth and death rates, $\beta(t)$ represents the time-varying transmission rate, and γ represents the recovery rate. The basic reproduction number $\mathcal{R}_0(t) = \beta(t)/(\gamma + \mu)$ is defined as the average number of secondary infections that a single infected individual would cause in a fully susceptible population at time t and measures the intrinsic transmissibility of a pathogen.

When we introduced the idea of pathogen resilience (Figure 2), we imposed sinusoidal changes to the transmission rate to account for seasonal transmission,

$$\beta(t) = b_1(1 + \theta \cos(2\pi(t - \phi)))\alpha(t), \quad (10)$$

where b_1 represents the baseline transmission rate, θ represents the seasonal amplitude, and ϕ represents the seasonal offset term. Here, we also introduced an extra multiplicative term $\alpha(t)$ to account for the impact of pandemic perturbations, where $\alpha(t) < 1$ indicates transmission reduction. Figure 2A and 2B were generated assuming $b_1 = 3 \times (365/7 + 1/50)/\text{years}$, $\theta = 0.2$, $\phi = 0$, $\mu = 1/50/\text{years}$, $\gamma = 365/7/\text{years}$, and $\delta = 1/2/\text{years}$. Specifically, $b_1 = 3 \times (365/7 + 1/50)/\text{years}$ implies $\mathcal{R}_0 = 3$, where $(365/7 + 1/50)/\text{years}$ represent the rate of recovery. In Figure 2A, we imposed a 50% transmission reduction for 6 months from 2020:

$$\alpha(t) = \begin{cases} 0.5 & 2020 \leq t < 2020.5 \\ 1 & \text{otherwise} \end{cases} \quad (11)$$

In Figure 2B, we imposed a 50% transmission reduction for 6 months from 2020 and a permanent 10% reduction onward:

$$\alpha(t) = \begin{cases} 1 & t < 2020 \\ 0.5 & 2020 \leq t < 2020.5 \\ 0.9 & 2020.5 \leq t \end{cases} \quad (12)$$

In both scenarios, we simulated the SIRS model from the same initial conditions ($S(0) = 1/\mathcal{R}_0$, $I(0) = 1 \times 10^{-6}$, and $R(0) = 1 - S(0) - I(0)$) from 1900 until 2030.

When we include importation in Supplementary Figures S2 and S3, we model the force of infection as $\beta(t)(I + \omega)$ instead and vary ω between 10^{-7} and 10^{-4} per day. Throughout the paper, all deterministic models were solved using the `lsoda` solver from the `deSolve` package [52] in R [53].

To measure the empirical resilience of the SIRS model (Figure 2C and 2F), we computed the normalized distance between post-intervention susceptible and logged infected proportions and their corresponding unperturbed values at the same time:

$$\sqrt{\left(\frac{S(t) - S_{\text{unperturbed}}(t)}{\sigma_S}\right)^2 + \left(\frac{\log I(t) - \log I_{\text{unperturbed}}(t)}{\sigma_{\log I}}\right)^2}, \quad (13)$$

where σ_S and $\sigma_{\log I}$ represent the standard deviation in the unperturbed susceptible and logged infected proportions. The unperturbed values were obtained by simulating the same SIRS model without pandemic perturbations ($\alpha = 1$). We normalized the differences in susceptible and logged infected proportions to allow both quantities to equally contribute to the changes in distance from the attractor. We used logged prevalence, instead of absolute prevalence, in order to capture epidemic dynamics in deep troughs during the intervention period. In Supplementary Materials, we also compared how the degree of seasonal transmission affects empirical resilience by varying θ from 0 to 0.4; when we assumed no seasonality ($\theta = 0$), we did not normalize the distance because the standard deviation of pre-intervention dynamics are zero.

We used the SIRS model to understand how underlying epidemiological parameters affect pathogen resilience and determine the relationship to underlying susceptible host dynamics. For the simple SIRS model without seasonal transmission ($\theta = 0$), the intrinsic resilience equals

$$-\text{Re}(\lambda) = \frac{\delta + \beta I^* + \mu}{2}. \quad (14)$$

Here, I^* represents the prevalence at endemic equilibrium:

$$I^* = \frac{(\delta + \mu)(\beta - (\gamma + \mu))}{\beta(\delta + \gamma + \mu)}. \quad (15)$$

The susceptible replenishment rate is given by

$$S_{\text{replenish}} = \frac{\mu + \delta R}{S^*}, \quad (16)$$

where $S^* = 1/\mathcal{R}_0$ represents the equilibrium proportion of susceptible individuals. We varied the basic reproduction number $\mathcal{R}_0$ between 1.1 to 6 and the average duration of immunity $1/\delta$ between 2 to 4 years, and computed these two quantities. In doing so, we fixed all other parameters: $\mu = 1/80/\text{years}$ and $\gamma = 365/7/\text{years}$. When infection provides a life-long immunity ($\delta = 0$), the model collapses to the SIR model

683 and the intrinsic resilience is exactly proportional to the susceptible replenishment
684 rate:

$$-\text{Re}(\lambda) = \frac{\mu}{2S^*} = \frac{S_{\text{replenish}}}{2}. \quad (17)$$

685 Finally, we used a seasonally unforced stochastic SIRS model without demog-
686 raphy to understand how pathogen resilience affects sensitivity of the system to
687 demographic stochasticity (see Supplementary Text for the details of the stochas-
688 tic SIRS model). By varying the basic reproduction number $\mathcal{R}_0$ between 2 to 20
689 and the average duration of immunity $1/\delta$ between 1 to 40 years, we ran the SIRS
690 model for 100 years and computed the epidemic amplitude, which we defined as
691 $(\max I - \min I)/(2\bar{I})$. Each simulation began from the equilibrium, and we trun-
692 cated the initial 25 years before computing the epidemic amplitude. In doing so,
693 we assumed $\gamma = 365/7/\text{years}$ and fixed the population size to 1 billion to prevent
694 any fadeouts. We also considered a seasonally forced stochastic SIRS model without
695 demography, assuming an amplitude of seasonal forcing of 0.04; in this case, we com-
696 puted the relative epidemic amplitude by comparing the deterministic and stochastic
697 trajectories (Supplementary Materials).

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702 Data availability

703 All data and code are stored in a publicly available GitHub repository (<https://github.com/cobeylab/return-time>).
704

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